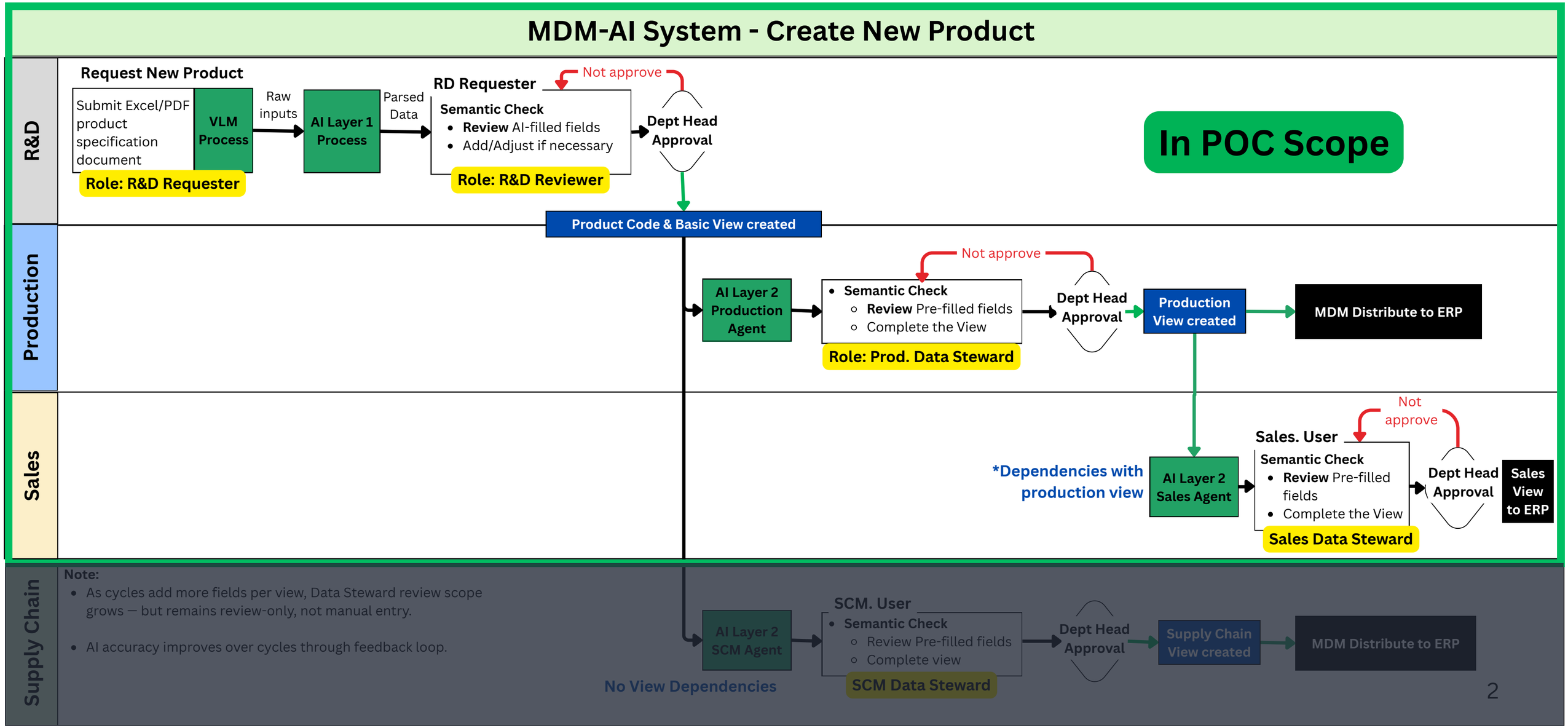
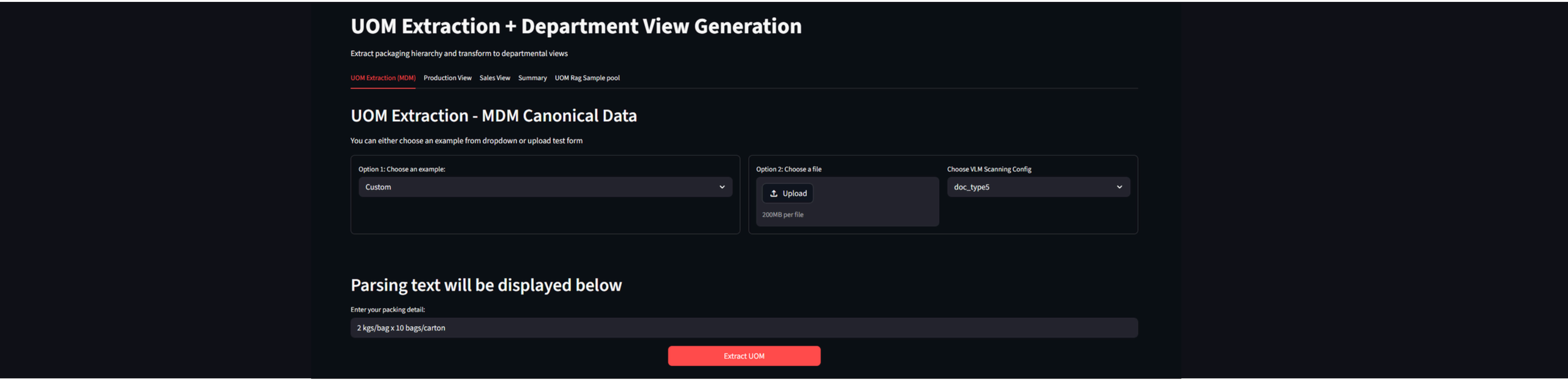
AI-MDM Pipeline POC - Streamlit Interface

Unit of Measurement Extraction & Transformation

POC automates canonical extraction and generates tailored department views



Four-stage pipeline extracts, transforms, and tracks UOM data with human oversight



Input

- **Direct text entry** OR **VLM** document scanning
- Handles nested structures (e.g., bag→carton)

MDM Cononical

- 15 fields extracted via **Tier 3 LLM+RAG**
- Per-field **confidence scoring**

Production View

- 2 fields derived via T**ier 2: Rule-based logic**
- **Manufacturing-specific** semantics
- **Conditional mapping** from canonical data

Sales View

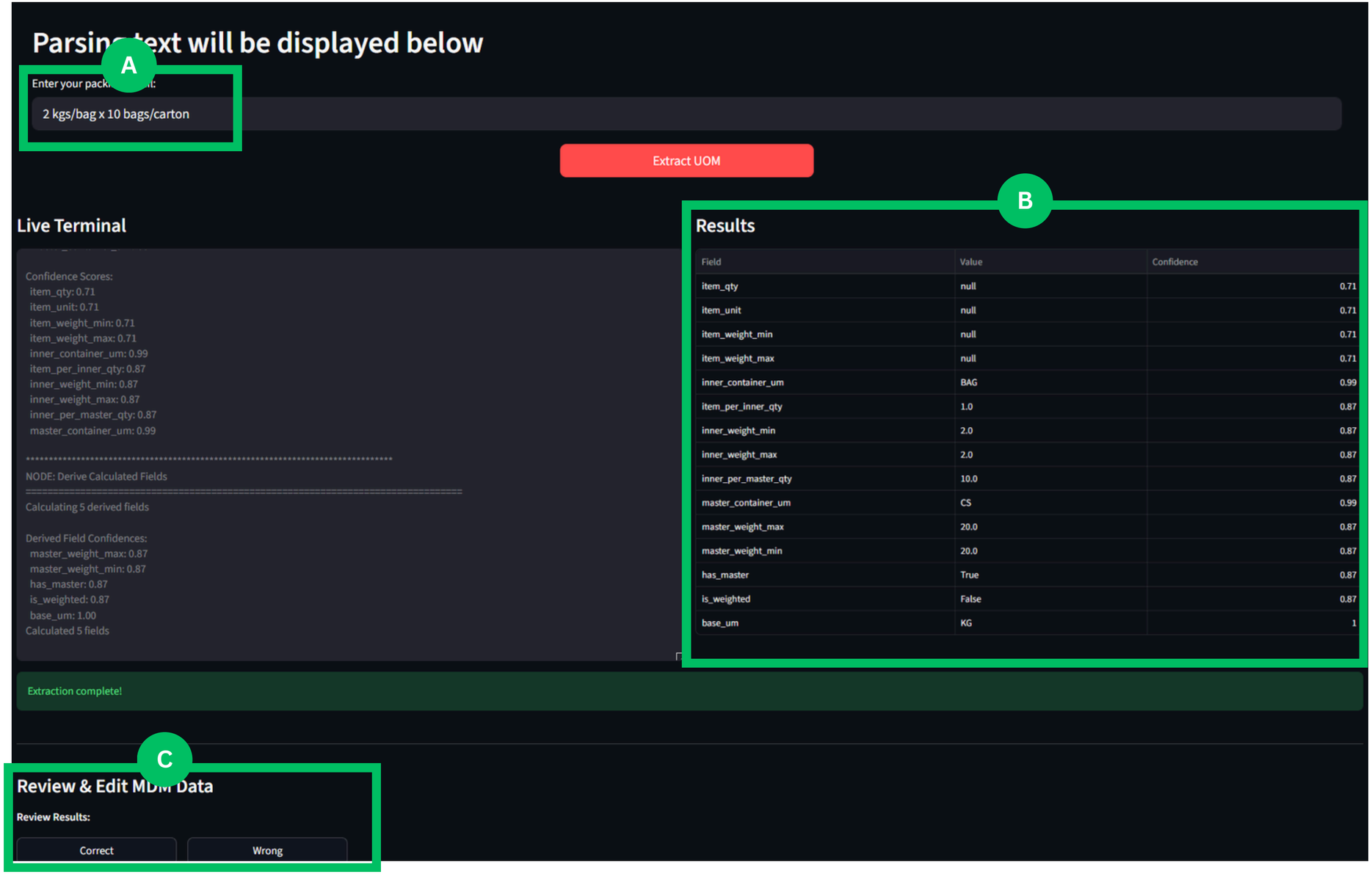
- 2 fields derived via **tier 1: direct mapping**
- **Depends on Production** View output
- Enforces **multi-hop lineage** consistency

Summary & Lineage

- All layers visible
- Metadata
- Field Lineage
- *Will be utilize in data catalog (Openmetadata)

Demo Shows: One input → MDM Canonical extraction → Tailored department views

LLM extracts 15 canonical fields from nested UOM text



A: Input

- Direct text entry OR VLM document scanning
- Handles nested structures
- Mixed Thai/English text supported

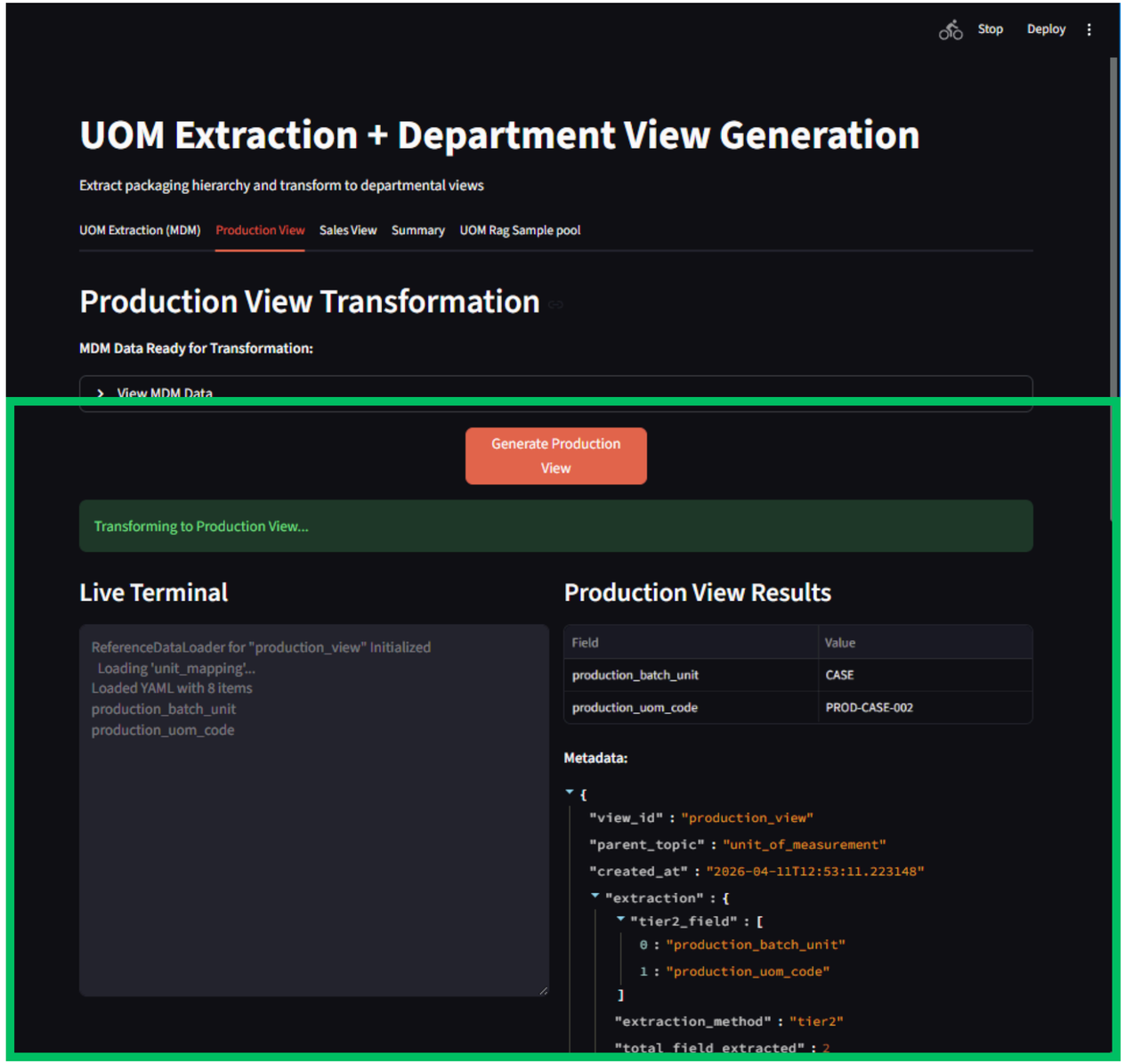
B: Extraction Results

- 15 canonical fields extracted automatically
- **Each field includes:**
  - Field name (e.g., master_container_um)
  - Extracted value (e.g., CS)
  - Confidence score (e.g., 0.99)

C: Human Review Gate

- **Review Results** section with two options:
  - **Correct** → Approve extraction as-is
  - **Wrong** → Opens inline editor for corrections
- All extractions **require explicit human approval**
- No auto-save without review

Production view derives 2 manufacturing-specific fields from canonical data via conditional logic



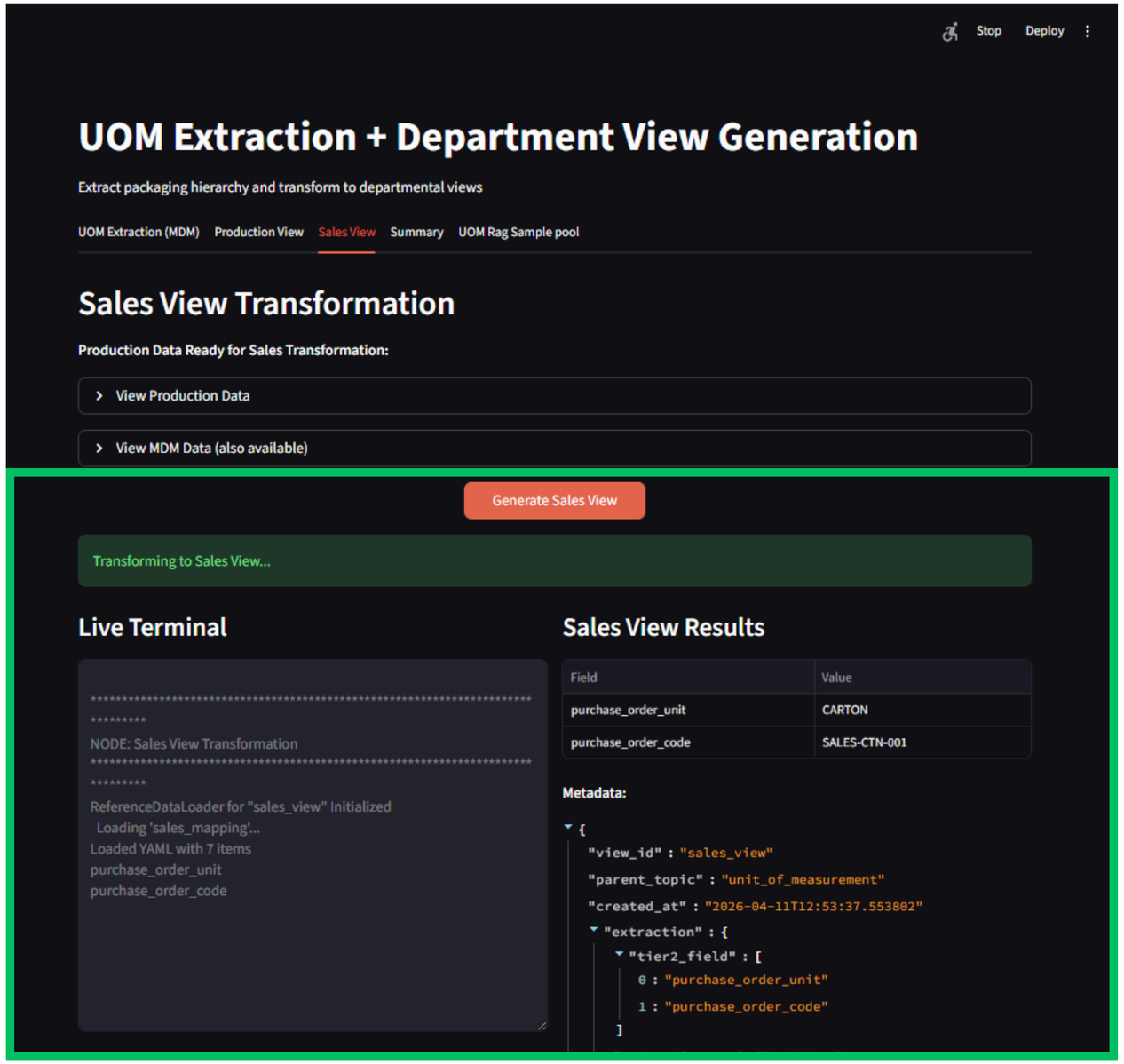
Context: What Production Needed

- Manufacturing **requires batch-level units for production planning**
- Standard container units from MDM Canonical Data (CS, CTN) don't match factory floor terminology
- Production codes **need PROD- prefix** for ERP integration

Transformation Overview

- **Fields Created:**
  - *production_batch_unit:* "CASE"
  - *production_uom_code*: "PROD-CASE-002"
- **Source: MDM canonical:**
  - *master_container_um*
  - *has_master*
  - *inner_container_um*
- **Method:**
  - Tier 2: Rule-based logic
  - YAML config defines conditional mapping
  - Deterministic transformation

Sales view depends on Production output, enforcing consistent multi-hop lineage



Context: What Sales Needed

- Sales department requires POS-friendly ordering units
- Purchase orders use customer-facing terminology (CARTON, not CASE)
- Sales codes need **SALES- prefix** for order management system
- **Must align with Production** to prevent inventory mismatches

Transformation Overview

- **Fields Created:**
  - *purchase_order_unit:* "CARTON"
  - *purchase_order_code:* "SALES-CTN-001"

- **Source: Production View**
  - Production view: *production_batch_unit*
  - Blocks generation until Production completes

- **Method:**
  - Tier 2: Rule-based mapping
  - YAML lookup table transforms Production units to Sales units
  - Enforces chain: Sales ← Production ← MDM

Three-column summary displays status, data, and metadata for governance validation

Extracted/Transformed Data Summary

UOM Extraction (MDM) | Production View | Sales View | Summary | UOM Rag Sample pool

Summary

MDM Canonical Data

Canonical Data


```
{
  "item_qty" : NULL
  "item_unit" : NULL
  "item_weight_min" : NULL
  "item_weight_max" : NULL
  "inner_container_um" : "BAG"
  "item_per_inner_qty" : 1
  "inner_weight_min" : 2
  "inner_weight_max" : 2
  "inner_per_master_qty" : 10
  "master_container_um" : "CS"
  "master_weight_max" : 20
  "master_weight_min" : 20
  "has_master" : true
  "is_weighted" : false
  "base_um" : "KG"
}
```


Production Data

Production Data


```
{
  "production_batch_unit" :
  "CASE"
  "production_uom_code" :
  "PROD-CASE-002"
}
```


Sales Data

Sales Data


```
{
  "purchase_order_unit" :
  "CARTON"
  "purchase_order_code" :
  "SALES-CTN-001"
}
```


Metadata

MDM Canonical Metadata

Metadata


```
{
  "record_identity" : {
    "record_id" :
    "UOM_20260411_125302"
    "topic" :
    "Unit of Measurement"
    "created_time" :
    "2026-04-11T12:53:02.847266"
    "record_version" : 1
  }
  "source" : {
    "source_system" :
    "rd_spec_sheet"
    "source_type" : "excel_form"
    "source_field" :
    "packing_detail"
    "source_extraction_method" :
    "vlm_qwen3-vl:8b"
  }
  "extraction" : {
    "tier3_fields" : [
      0 : "item_qty"
      1 : "item_unit"
      2 : "item_weight_min"
      3 : "item_weight_max"
      4 : "inner_container_um"
      5 : "item_per_inner_qty"
      6 : "inner_weight_min"
      7 : "inner_weight_max"
      8 : "inner_per_master_qty"
      9 : "master_container_um"
    ]
```


Production Metadata

Production Metadata


```
{
  "view_id" : "production_view"
  "parent_topic" :
  "unit_of_measurement"
  "created_at" :
  "2026-04-11T12:53:11.223148"
  "extraction" : {
    "tier2_field" : [
      0 :
      "production_batch_unit"
      1 : "production_uom_code"
    ]
    "extraction_method" : "tier2"
    "total_field_extracted" : 2
  }
  "lineage" : {
    "field_dependencies" : {
      "production_batch_unit" :
      {
        "depends_on" : [
          0 : "has_master"
          1 :
          "master_container_
          um"
          2 :
          "inner_container_u
          m"
        ]
      }
      "production_uom_code" :
      {
        "depends_on" : [
          0 :
```


Sales Metadata

Sales Metadata


```
{
  "view_id" : "sales_view"
  "parent_topic" :
  "unit_of_measurement"
  "created_at" :
  "2026-04-11T12:53:37.553802"
  "extraction" : {
    "tier2_field" : [
      0 : "purchase_order_unit"
      1 : "purchase_order_code"
    ]
    "extraction_method" : "tier1"
    "total_field_extracted" : 2
  }
  "lineage" : {
    "field_dependencies" : {
      "purchase_order_unit" :
      {
        "depends_on" : [
          0 :
          "production_batch_
          unit"
        ]
      }
      "purchase_order_code" :
      {
        "depends_on" : [
          0 :
          "production_batch_
          unit"
        ]
      }
    }
```

Sample execution proves end-to-end pipeline reliability

Stage	Field Count	Field Examples	Transformation
Input	**1**	• "2 kgs/bag x 10 bags/carton"	User input
MDM Canonical	**15**	• master_container_um: "CS" • is_weighted: False	Tier 3 -LLM RAG-Enhanced
Production	**2**	• production_batch_unit: "CASE" • production_uom_code: “PROD-CASE-002”	Tier 2 - Rule Based Logic
Sales	**2**	• purchase_order_unit: "CARTON" • purchase_order_code: “SALES-CTN-001”	Tier 2 - Rule Based Logic